Model Architecture: Qwen3.5-0.8B

Generated by sg_modelload

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1 Model Overview

Model name	Qwen3.5-0.8B
Transformers class	`Qwen3_5ForConditionalGeneration`
Path	`/models/memorization/Qwen3.5-0.8B`
Total parameters	852,985,920 (852.99M)
Checkpoint-only tensors (not in total)	20,452,864
Weight dtype	bfloat16
Hidden size d	1,024
Decoder layers L	24
Query heads H	8
Key/value heads H_{kv}	2
Head dim d_h	256
MLP width d_{ff}	3,584
DeltaNet q/k heads	16
DeltaNet v heads	16
DeltaNet q/k head dim	128
DeltaNet v head dim	128
DeltaNet conv kernel	4
Vocabulary V	248,320
Max context length	262,144
RMSNorm ϵ	1e-06
Tied input/output embeddings	True
Activation	SiLU
RoPE base θ	10,000,000
Vision encoder: layers	12
hidden size / heads	768 / 12

How to read this document. Section 2 writes the whole network as one composed function. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Notation

B, T	batch size, sequence length (tokens)
d	hidden size (1024)
V	vocabulary size (248,320)
H, H_{kv}, d_h	query heads, key/value heads, size of one head
W_*, b_*	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
RMSNorm(x)	$\gamma \odot x / \sqrt{\text{mean}(x^2) + \epsilon}$ (learned scale γ)
LN(x)	LayerNorm: $\gamma \odot (x - \mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 The Model as a Function

Input: token ids $x \in \{0, \ldots, V-1\}^{B \times T}$. Output: next-token logits $y \in \mathbb{R}^{B \times T \times V}$. The 24 decoder layers are written with composition powers, so a block that repeats n times appears once as $\mathcal{B}^{\circ n}$.

$$h_0 = \text{Merge}\Big(\text{Embed}(x),\ \text{Proj}_{\text{img}}\big(\text{Enc}_{\text{V}}(\text{img})\big)\Big)$$

$$h_L = \Big(\big(\mathcal{B}_{\text{F}} \circ \mathcal{B}_{\text{L}}^{\circ 3}\big)^{\circ 6}\Big)(h_0)$$

$$y = W_E^{\top}\ \text{RMSNorm}_{\text{final}}(h_L)$$

Merge puts the projected image/audio vectors at the placeholder-token positions of the text embedding sequence. The encoders are themselves repeated stacks:

$$\text{Enc}_{\text{V}} = \text{Post} \circ \mathcal{B}_{\text{V}}^{\circ 12} \circ \text{PatchEmbed}$$

2.1 Layer pattern

Symbol	Block type	Count	Params / block	Layer indices
$\mathcal{B}_{\mathrm{L}}$	Linear-attention (Gated DeltaNet) block	18	21,555,360	0–2, 4–6, 8–10, 12–14, 16–18, 20–22
$\mathcal{B}_{\mathrm{F}}$	Full (global) attention block	6	18,352,640	3, 7, …, 23 (every 4th)
$\mathcal{B}_{\mathrm{V}}$	Vision encoder block	12	7,087,872	0–11

Language-model decoder (24 layers). layers 0–23: $[\mathcal{B}_{\mathrm{L}}\times 3, \mathcal{B}_{\mathrm{F}}]$ repeated $6\times$.

Vision encoder (12 layers). layers 0–11: $[\mathcal{B}_{\mathrm{V}}]$ repeated $12\times$.

2.2 Block functions

$\mathcal{B}_{\mathrm{L}}$ — **Linear-attention (Gated DeltaNet) block.**

$$
\begin{aligned}
a &= h + \mathrm{Mixer}(\mathrm{RMSNorm}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
[q,\, k,\, v] &= \mathrm{SiLU}\big(\,\mathrm{CausalConv1d}(W_{qkv}\,u)\big) && \text{short causal conv (kernel 4) over time}\\
z &= W_z\,u && \text{output gate}\\
\beta_t &= \sigma(W_b\,u_t) && \text{write strength in (0,1)}\\
g_t &= -e^{A_{\log}}\cdot \mathrm{softplus}(W_a\,u_t + b_{dt}) && \text{decay } (\le 0)\text{: how fast memory fades}\\
q_t,\, k_t &= \mathrm{L2Norm}(q_t),\ \mathrm{L2Norm}(k_t)\\
S_t &= e^{g_t}\,S_{t-1}\big(I - \beta_t k_t k_t^{\top}\big) + \beta_t\, v_t k_t^{\top} && \text{fixed-size memory per head (delta rule)}\\
o_t &= S_t\,q_t && \text{read memory with the query}\\
\mathrm{Mixer}(u) &= W_{\mathrm{out}}\big(\,\mathrm{RMSNorm}(o)\odot \mathrm{SiLU}(z)\big)\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\,\mathrm{SiLU}(W_{\mathrm{gate}}\,u)\odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width 3584}
\end{aligned}
$$

$\mathcal{B}_{\mathrm{F}}$ — **Full (global) attention block.**

$$
\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{RMSNorm}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
[Q,\, G] &= W_Q\,u && \text{query and an output gate}\\
K,\, V &= W_K\,u,\quad W_V\,u && H_{kv}=2 \text{ key/value heads}\\
Q,\, K &= \mathrm{RMSNorm}_q(Q),\ \mathrm{RMSNorm}_k(K) && \text{per-head normalisation}\\
Q,\, K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\,Q_h K_{g(h)}^{\top} + M && \text{mask } M\text{: causal}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\,V_{g(h)}\\
\mathrm{Attn}(u) &= W_O\big([O_1;\ldots;O_H]\odot\sigma(G)\big)\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\,\mathrm{SiLU}(W_{\mathrm{gate}}\,u)\odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width 3584}
\end{aligned}
$$

$\mathcal{B}_{\mathrm{V}}$ — **Vision encoder block.**

$$
\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{LN}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{LN}_2(a)) && \text{pre-norm + residual}\\
[Q,\, K,\, V] &= W_{qkv}\,u + b_{qkv} && \text{one fused projection}\\
Q,\, K &= \mathrm{RoPE}_{2D}(Q),\ \mathrm{RoPE}_{2D}(K) && \text{rotate by (row, column) patch position}\\
S_h &= s\,Q_h K_h^{\top} + M && \text{mask } M\text{: bidirectional}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\,V_h\\
\mathrm{Attn}(u) &= W_O\,[O_1;\ldots;O_H] + b_O\\
\mathrm{FFN}(u) &= W_2\ \mathrm{GELU}_{\tanh}(W_1 u + b_1) + b_2 && \text{MLP, width 3072}
\end{aligned}
$$

3 Forward Pass in Pseudocode

Algorithm 1 Qwen3.5-0.8B: full forward pass

Require: token ids x of shape $B \times T$; optional images / audio

$h \leftarrow \mathrm{Embed}(x)$ ▷ look up one vector per token: $B{\times}T{\times}1024$
if image input present **then**
  $e \leftarrow \mathrm{Enc}_{\mathrm{V}}(\text{image})$ ▷ vision encoder: 12 layers, Algorithm 2
  $h[\text{image token slots}] \leftarrow \mathrm{Proj}(e)$ ▷ project into text width 1024
end if
for $r \leftarrow 0$ **to** 5 **do** ▷ layers 0–23: pattern of 4 repeated 6 times
  for $j \leftarrow 1$ **to** 3 **do**
    $h \leftarrow \mathcal{B}_{\mathrm{L}}(h)$ ▷ Linear-attention (Gated DeltaNet) block
  end for
  $h \leftarrow \mathcal{B}_{\mathrm{F}}(h)$ ▷ Full (global) attention block
end for
$h \leftarrow \mathrm{RMSNorm}_{\mathrm{final}}(h)$
$y \leftarrow h\,W_E^{\top}$ ▷ logits $B{\times}T{\times}248,320$, embedding matrix re-used
return y ▷ $\mathrm{softmax}(y)$ = next-token probabilities

Algorithm 2 Vision encoder

$h \leftarrow \mathrm{PatchEmbed}(\text{pixels}) + \text{position embedding}$ ▷ one vector per image patch
for $\ell \leftarrow 0$ **to** 11 **do**
  $h \leftarrow \mathcal{B}_{\mathrm{V}}(h)$
end for
$h \leftarrow \mathrm{Post}(h)$ ▷ final norm / pooling / merging, see Table 2
return h

Algorithm 3 Block $\mathcal{B}_{\mathrm{L}}(h)$ — Linear-attention (Gated DeltaNet) block

Require: h of shape $B \times T \times 1024$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: Gated DeltaNet (linear attention, fixed-size memory)
$q, k, v \leftarrow \mathrm{split}(\mathrm{SiLU}(\mathrm{CausalConv1d}(uW_{qkv}^{\top})))$ ▷ 16 q/k heads of 128, 16 v heads of 128
$z \leftarrow uW_z^{\top};\ \beta \leftarrow \sigma(uW_b^{\top});\ g \leftarrow -e^{A_{\log}}\,\mathrm{softplus}(uW_a^{\top} + b_{dt})$ ▷ gate, write strength, decay
for $t \leftarrow 1$ **to** T **do** ▷ conceptually sequential; computed in chunks
  $S \leftarrow e^{g_t} S\,(I - \beta_t k_t k_t^{\top}) + \beta_t v_t k_t^{\top}$ ▷ forget a bit, overwrite what k_t points at
  $o_t \leftarrow S\,q_t$ ▷ read
end for
$m \leftarrow \mathrm{RMSNorm}(o) \odot \mathrm{SiLU}(z)\ W_{\mathrm{out}}^{\top}$ ▷ back to $B{\times}T{\times}1024$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{SiLU}(uW_{\mathrm{gate}}^{\top}) \odot uW_{\mathrm{up}}^{\top}\big)W_{\mathrm{down}}^{\top}$ ▷ expand to 3584, gate, shrink back to 1024
$h \leftarrow r + m$ ▷ add residual
return h

Algorithm 4 Block $\mathcal{B}_{\mathrm{F}}(h)$ — Full (global) attention block

Require: h of shape $B \times T \times 1024$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}8{\times}256$ (8 query heads); second half is a gate G
$K \leftarrow uW_K^\top;\ V \leftarrow uW_V^\top$ ▷ $B{\times}T{\times}2{\times}256$ (2 key/value heads, each shared by 4 query heads)
$Q \leftarrow \mathrm{RMSNorm}(Q);\ K \leftarrow \mathrm{RMSNorm}(K)$ ▷ normalise each head vector (stabilises training)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: causal: a token sees only earlier tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}8{\times}256$
$O \leftarrow O \odot \sigma(G)$ ▷ output gate decides how much of each head to keep
$m \leftarrow O\,W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}1024$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{SiLU}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big)W_{\mathrm{down}}^\top$ ▷ expand to 3584, gate, shrink back to 1024
$h \leftarrow r + m$ ▷ add residual
return h

Algorithm 5 Block $\mathcal{B}_{\mathrm{V}}(h)$ — Vision encoder block

Require: h of shape $B \times T \times 768$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{LN}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q, K, V \leftarrow \mathrm{split}(uW_{qkv}^\top + b)$ ▷ each $B{\times}T{\times}12{\times}64$
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ 2-D rotary position (row, column)
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: every patch/frame sees every other
$O \leftarrow PV$ ▷ $B{\times}T{\times}12{\times}64$
$m \leftarrow O\,W_O^\top + b_O$ ▷ concatenate heads, project back to $B{\times}T{\times}768$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{LN}(h)$
$m \leftarrow \mathrm{GELU}_{\tanh}(uW_1^\top + b_1)W_2^\top + b_2$ ▷ expand to 3072, shrink back to 768
$h \leftarrow r + m$ ▷ add residual
return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 2: Weights that appear once

Weight	Type	Shape	Params
`model.visual.patch_embed.proj.weight`	Conv3d	$768 \times 3 \times 2 \times 16 \times 16$	1,179,648
`model.visual.patch_embed.proj.bias`	Conv3d	768	768
`model.visual.pos_embed.weight`	Embedding	2304×768	1,769,472
`model.visual.merger.norm.weight`	LayerNorm	768	768
`model.visual.merger.norm.bias`	LayerNorm	768	768
`model.visual.merger.linear_fc1.weight`	Linear	3072×3072	9,437,184
`model.visual.merger.linear_fc1.bias`	Linear	3072	3,072
`model.visual.merger.linear_fc2.weight`	Linear	1024×3072	3,145,728
`model.visual.merger.linear_fc2.bias`	Linear	1024	1,024
`model.language_model.embed_tokens.weight`	Embedding	248320×1024	254,279,680
`model.language_model.norm.weight`	Qwen3_5RMSNorm	1024	1,024

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Weight	Type	Shape	Params
`lm_head.weight` (tied to `model.language_model.embed_tokens.weight`)	Linear	248320×1024	0 (shared)

4.2 Language-model decoder: `model.language_model.layers`

Table 3: $\mathcal{B}_{\mathrm{L}}$: Linear-attention (Gated DeltaNet) block — layers 0–2, 4–6, 8–10, 12–14, 16–18, 20–22; weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`linear_attn.dt_bias`	Qwen3_5GatedDeltaNet	16	16
`linear_attn.A_log`	Qwen3_5GatedDeltaNet	16	16
`linear_attn.conv1d.weight`	Conv1d	$6144 \times 1 \times 4$	24,576
`linear_attn.norm.weight`	Qwen3_5RMSNormGated	128	128
`linear_attn.out_proj.weight`	Linear	1024×2048	2,097,152
`linear_attn.in_proj_qkv.weight`	Linear	6144×1024	6,291,456
`linear_attn.in_proj_z.weight`	Linear	2048×1024	2,097,152
`linear_attn.in_proj_b.weight`	Linear	16×1024	16,384
`linear_attn.in_proj_a.weight`	Linear	16×1024	16,384
`mlp.gate_proj.weight`	Linear	3584×1024	3,670,016
`mlp.up_proj.weight`	Linear	3584×1024	3,670,016
`mlp.down_proj.weight`	Linear	1024×3584	3,670,016
`input_layernorm.weight`	Qwen3_5RMSNorm	1024	1,024
`post_attention_layernorm.weight`	Qwen3_5RMSNorm	1024	1,024
Per block			**21,555,360**
× 18 blocks			**387,996,480**

Table 4: $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block — layers 3, 7, …, 23 (every 4th); weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	4096×1024	4,194,304
`self_attn.k_proj.weight`	Linear	512×1024	524,288
`self_attn.v_proj.weight`	Linear	512×1024	524,288
`self_attn.o_proj.weight`	Linear	1024×2048	2,097,152
`self_attn.q_norm.weight`	Qwen3_5RMSNorm	256	256
`self_attn.k_norm.weight`	Qwen3_5RMSNorm	256	256
`mlp.gate_proj.weight`	Linear	3584×1024	3,670,016
`mlp.up_proj.weight`	Linear	3584×1024	3,670,016
`mlp.down_proj.weight`	Linear	1024×3584	3,670,016
`input_layernorm.weight`	Qwen3_5RMSNorm	1024	1,024
`post_attention_layernorm.weight`	Qwen3_5RMSNorm	1024	1,024
Per block			**18,352,640**
× 6 blocks			**110,115,840**

4.3 Vision encoder: `model.visual.blocks`

Table 5: $\mathcal{B}_V$: Vision encoder block — layers 0–11; weight names are relative to `model.visual.blocks.`ℓ

Weight (relative)	Type	Shape	Params
`norm1.weight`	LayerNorm	768	768
`norm1.bias`	LayerNorm	768	768
`norm2.weight`	LayerNorm	768	768
`norm2.bias`	LayerNorm	768	768
`attn.qkv.weight`	Linear	2304×768	1,769,472
`attn.qkv.bias`	Linear	2304	2,304
`attn.proj.weight`	Linear	768×768	589,824
`attn.proj.bias`	Linear	768	768
`mlp.linear_fc1.weight`	Linear	3072×768	2,359,296
`mlp.linear_fc1.bias`	Linear	3072	3,072
`mlp.linear_fc2.weight`	Linear	768×3072	2,359,296
`mlp.linear_fc2.bias`	Linear	768	768
Per block			**7,087,872**
× 12 blocks			**85,054,464**

4.4 Checkpoint-only modules

Multi-token-prediction (MTP) head (`mtp.*`). An extra one-layer transformer that predicts the token after next. It concatenates RMSNorm(embedding of token $t{+}1$) with RMSNorm(final hidden state at t), projects $2d \to d$ (`fc`), runs one full-attention decoder layer, then a norm and the shared LM head. Serving engines (vLLM/SGLang) use it for speculative decoding. The transformers class does not instantiate it, so it is not counted in the total above.

Tensor	Shape	Dtype	Params
`mtp.fc.weight`	1024×2048	bf16	2,097,152
`mtp.layers.0.input_layernorm.weight`	1024	bf16	1,024
`mtp.layers.0.mlp.down_proj.weight`	1024×3584	bf16	3,670,016
`mtp.layers.0.mlp.gate_proj.weight`	3584×1024	bf16	3,670,016
`mtp.layers.0.mlp.up_proj.weight`	3584×1024	bf16	3,670,016
`mtp.layers.0.post_attention_layernorm.weight`	1024	bf16	1,024
`mtp.layers.0.self_attn.k_norm.weight`	256	bf16	256
`mtp.layers.0.self_attn.k_proj.weight`	512×1024	bf16	524,288
`mtp.layers.0.self_attn.o_proj.weight`	1024×2048	bf16	2,097,152
`mtp.layers.0.self_attn.q_norm.weight`	256	bf16	256
`mtp.layers.0.self_attn.q_proj.weight`	4096×1024	bf16	4,194,304
`mtp.layers.0.self_attn.v_proj.weight`	512×1024	bf16	524,288
`mtp.norm.weight`	1024	bf16	1,024
`mtp.pre_fc_norm_embedding.weight`	1024	bf16	1,024
`mtp.pre_fc_norm_hidden.weight`	1024	bf16	1,024

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
LANGUAGE MODEL	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?". Here it also outputs an equally sized gate for the attention output (hence the doubled width).
Query norm (q_norm) `self_attn.q_norm`	RMSNorm applied to every query head vector before RoPE. Keeps attention logits in a sane range (QK-norm).

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Layer (appears as)	What it does
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Key norm (k_norm) `self_attn.k_norm`	RMSNorm applied to every key head vector before RoPE.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
DeltaNet output projection `linear_attn.out_proj`	Mixes the heads back into one hidden-size vector.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
DeltaNet Q/K/V projection `linear_attn.in_proj_qkv`	Produces queries, keys and values for the linear-attention layer in one matrix.
DeltaNet output-gate projection `linear_attn.in_proj_z`	Produces gate z. The normalised output is multiplied by SiLU(z).
DeltaNet write-strength projection `linear_attn.in_proj_b`	Produces $\beta = \sigma(\cdot)$ per head: how strongly the current token overwrites the memory.
DeltaNet decay projection `linear_attn.in_proj_a`	Produces the input-dependent forgetting rate (with A_log and dt_bias).
Short causal convolution `linear_attn.conv1d`	Depthwise 1-D convolution over the last 4 tokens of Q/K/V. Gives cheap local context before the recurrent memory update.
A_log (decay base) `layers.linear_attn`	Per-head learned log decay rate. $-e^{A_{\log}}$ sets how quickly old memory fades.
dt_bias (decay bias) `layers.linear_attn`	Per-head bias added before the softplus that controls the decay step.
Gated output RMSNorm `linear_attn.norm`	Normalises each head's memory read-out and multiplies it by SiLU(z).
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	Despite the name, this is the pre-MLP norm: RMSNorm on the residual stream after attention, feeding the MLP.
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.
VISION ENCODER	
3-D patch embedding (Conv3d) `patch_embed.proj`	Cuts images/videos into 2-frame × 16 × 16-pixel cubes and maps each cube to one vector.
Position embedding `visual.pos_embed`	Learned position vectors, interpolated to the image's patch grid.
Pre-attention LayerNorm `blocks.norm1`	LayerNorm before vision self-attention.
Pre-MLP LayerNorm `blocks.norm2`	LayerNorm before the vision MLP.
Fused Q/K/V projection `attn.qkv`	One matrix that produces queries, keys and values at once (split afterwards).
Attention output projection `attn.proj`	Mixes the concatenated head outputs back to the model width.

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Layer (appears as)	What it does
MLP up-projection (fc1) `mlp.linear_fc1`	Expands each vector to the wider MLP size before the GELU non-linearity.
MLP down-projection (fc2) `mlp.linear_fc2`	Shrinks back to the model width.
Merger LayerNorm `merger.norm`	Normalises patch features before 2×2 neighbouring patches are merged.
Merger MLP layer 1 `merger.linear_fc1`	Takes 4 neighbouring patches concatenated (2×2 merge, so 4× fewer image tokens) and mixes them.
Merger MLP layer 2 (to text width) `merger.linear_fc2`	Projects merged patches into the language model's hidden size.

6 Parameter Count

Table 8: Parameters by component

Component	Parameters	% of total
MLP / MoE	264,241,152	31.0%
Token embedding	254,279,680	29.8%
Attention / token mixer	233,822,016	27.4%
Vision encoder	100,592,896	11.8%
Norms	50,176	0.0%
Total	**852,985,920**	**100.0%**

Table 9: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{\mathrm{L}}$ Linear-attention (Gated DeltaNet) block	18	21,555,360	387,996,480
$\mathcal{B}_{\mathrm{F}}$ Full (global) attention block	6	18,352,640	110,115,840
$\mathcal{B}_{\mathrm{V}}$ Vision encoder block	12	7,087,872	85,054,464
Modules outside stacks	1	–	269,819,136
Total			**852,985,920**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
`lm_head.weight` *(tied)*	Linear	248320 × 1024	bfloat16	254,279,680
`model`	Qwen3_5Model	–	–	–
`model.language_model`	Qwen3_5TextModel	–	–	–
`model.language_model.embed_tokens.weight`	Embedding	248320 × 1024	bfloat16	254,279,680
`model.language_model.layers.0`	Qwen3_5DecoderLayer	–	–	–
`model.language_model.layers.0.input_layernorm.weight`	Qwen3_5RMSNorm	1024	bfloat16	1,024
`model.language_model.layers.0.linear_attn.A_log`	Qwen3_5GatedDeltaNet	16	bfloat16	16
`model.language_model.layers.0.linear_attn`	Qwen3_5GatedDeltaNet	–	–	–
`model.language_model.layers.0.linear_attn.conv1d.weight`	Conv1d	6144 × 1 × 4	bfloat16	24,576
`model.language_model.layers.0.linear_attn.dt_bias`	Qwen3_5GatedDeltaNet	16	bfloat16	16
`model.language_model.layers.0.linear_attn.in_proj_a.weight`	Linear	16 × 1024	bfloat16	16,384
`model.language_model.layers.0.linear_attn.in_proj_b.weight`	Linear	16 × 1024	bfloat16	16,384
`model.language_model.layers.0.linear_attn.in_proj_qkv.weight`	Linear	6144 × 1024	bfloat16	6,291,456

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.0.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.0.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.0.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.0.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.0.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.0.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.0.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.0.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.1	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.1.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.1.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.1.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.1.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.1.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.1.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.1.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.1.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.1.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.1.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.1.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.1.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.1.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.1.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.1.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.1.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.2	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.2.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.2.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.2.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.2.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.2.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.2.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.2.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.2.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.2.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.2.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.2.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.2.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.2.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.2.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.2.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.2.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.3	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.3.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.3.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.3.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.3.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.3.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.3.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.3.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.3.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.3.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.3.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.3.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.3.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.language_model.layers.3.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.4	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.4.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.4.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.4.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.4.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.4.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.4.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.4.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.4.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.4.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.4.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.4.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.4.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.4.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.4.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.4.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.4.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.5	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.5.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.5.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.5.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.5.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.5.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.5.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.5.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.5.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.5.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.5.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.5.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.5.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.5.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.5.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.5.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.5.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.6	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.6.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.6.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.6.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.6.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.6.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.6.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.6.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.6.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.6.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.6.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.6.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.6.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.6.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.6.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.6.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.6.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.7	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.7.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.7.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.7.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.7.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.7.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.7.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.7.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.7.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.7.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.7.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.7.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.7.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.language_model.layers.7.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.8	Qwen3_5DecoderLayer	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.8.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.8.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.8.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.8.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.8.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.8.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.8.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.8.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.8.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.8.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.8.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.8.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.8.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.8.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.8.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.8.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.9	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.9.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.9.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.9.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.9.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.9.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.9.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.9.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.9.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.9.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.9.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.9.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.9.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.9.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.9.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.9.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.9.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.10	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.10.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.10.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.10.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.10.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.10.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.10.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.10.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.10.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.10.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.10.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.10.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.10.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.10.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.10.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.10.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.10.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.11	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.11.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.11.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.11.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.11.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.11.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.11.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.11.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.11.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.11.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.11.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.11.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.11.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.language_model.layers.11.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.12	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.12.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.12.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.12.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.12.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.12.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.12.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.12.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.12.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.12.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.12.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.12.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.12.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.12.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.12.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.12.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.12.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.13	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.13.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.13.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.13.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.13.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.13.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.13.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.13.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.13.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.13.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.13.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.13.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.13.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.13.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.13.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.13.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.13.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.14	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.14.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.14.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.14.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.14.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.14.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.14.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.14.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.14.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.14.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.14.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.14.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.14.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.14.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.14.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.14.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.14.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.15	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.15.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.15.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.15.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.15.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.15.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.15.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.15.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.15.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.15.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.15.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.15.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.15.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.language_model.layers.15.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.16	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.16.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.16.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.16.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.16.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.16.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.16.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.16.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.16.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.16.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.16.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.16.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.16.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.16.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.16.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.16.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.16.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.17	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.17.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.17.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.17.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.17.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.17.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.17.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.17.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.17.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.17.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.17.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.17.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.17.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.17.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.17.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.17.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.17.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.18	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.18.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.18.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.18.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.18.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.18.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.18.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.18.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.18.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.18.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.18.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.18.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.18.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.18.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.18.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.18.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.18.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.19	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.19.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.19.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.19.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.19.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.19.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.19.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.19.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.19.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.19.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.19.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.19.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.19.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.language_model.layers.19.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.20	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.20.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.20.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.20.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.20.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.20.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.20.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.20.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.20.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.20.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.20.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.20.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.20.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.20.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.20.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.20.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.20.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.21	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.21.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.21.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.21.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.21.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.21.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.21.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.21.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.21.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.21.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.21.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.21.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.21.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.21.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.21.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.21.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.21.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.22	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.22.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.22.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.22.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.22.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.22.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.22.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.22.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.22.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.22.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.22.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.22.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.22.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.22.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.22.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.22.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.22.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.23	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.23.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.23.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.23.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.23.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.23.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.23.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.23.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.23.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.23.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.23.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.23.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.23.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.language_model.layers.23.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.norm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.visual	Qwen3_5VisionModel	–	–	–
model.visual.blocks.0	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.0.attn	Qwen3_5VisionAttention	–	–	–
model.visual.blocks.0.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.0.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.0.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.0.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.0.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.0.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.0.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.0.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.0.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.0.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.0.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.0.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.0.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.1	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.1.attn	Qwen3_5VisionAttention	–	–	–
model.visual.blocks.1.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.1.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.1.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.1.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.1.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.1.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.1.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.1.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.1.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.1.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.1.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.1.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.1.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.2	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.2.attn	Qwen3_5VisionAttention	–	–	–
model.visual.blocks.2.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.2.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.2.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.2.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.2.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.2.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.2.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.2.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.2.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.2.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.2.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.2.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.2.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.3	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.3.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.3.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.3.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.3.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.3.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.3.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.3.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.3.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.3.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.3.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.3.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.3.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.3.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.3.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.4	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.4.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.4.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.4.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.4.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.4.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.4.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.4.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.4.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.4.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.4.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.4.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.4.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.4.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.4.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.5	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.5.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.5.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.5.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.5.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.5.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.5.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.5.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.5.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.5.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.5.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.5.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.5.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.5.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.5.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.6	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.6.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.6.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.6.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.6.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.6.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.6.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.6.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.6.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.6.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.6.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.6.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.6.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.6.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.6.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.7	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.7.attn	Qwen3_5Vision Attention	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.7.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.7.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.7.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.7.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.7.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.7.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.7.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.7.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.7.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.7.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.7.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.7.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.7.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.8	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.8.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.8.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.8.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.8.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.8.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.8.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.8.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.8.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.8.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.8.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.8.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.8.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.8.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.8.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.9	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.9.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.9.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.9.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.9.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.9.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.9.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.9.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.9.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.9.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.9.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.9.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.9.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.9.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.9.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.10	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.10.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.10.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.10.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.10.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.10.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.10.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.10.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.10.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.10.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.10.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.10.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.10.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.10.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.10.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.11	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.11.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.11.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.11.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.11.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.11.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.11.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.11.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.11.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.11.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.11.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.11.norm1.bias	LayerNorm	768	bfloat16	768

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.11.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.11.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.11.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.merger	Qwen3_5VisionPatch Merger	–	–	–
model.visual.merger.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.merger.linear_fc1.weight	Linear	3072 × 3072	bfloat16	9,437,184
model.visual.merger.linear_fc2.bias	Linear	1024	bfloat16	1,024
model.visual.merger.linear_fc2.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.visual.merger.norm.bias	LayerNorm	768	bfloat16	768
model.visual.merger.norm.weight	LayerNorm	768	bfloat16	768
model.visual.patch_embed	Qwen3_5VisionPatch Embed	–	–	–
model.visual.patch_embed.proj.bias	Conv3d	768	bfloat16	768
model.visual.patch_embed.proj.weight	Conv3d	768 × 3 × 2 × 16 × 16	bfloat16	1,179,648
model.visual.pos_embed.weight	Embedding	2304 × 768	bfloat16	1,769,472

Appendix B Per-Layer Parameter Count

Module	Params	Running total
model.language_model.embed_tokens	254,279,680	254,279,680
model.language_model.layers.0.input_layernorm	1,024	254,280,704
model.language_model.layers.0.linear_attn	32	254,280,736
model.language_model.layers.0.linear_attn.conv1d	24,576	254,305,312
model.language_model.layers.0.linear_attn.in_proj_a	16,384	254,321,696
model.language_model.layers.0.linear_attn.in_proj_b	16,384	254,338,080
model.language_model.layers.0.linear_attn. in_proj_qkv	6,291,456	260,629,536
model.language_model.layers.0.linear_attn.in_proj_z	2,097,152	262,726,688
model.language_model.layers.0.linear_attn.norm	128	262,726,816
model.language_model.layers.0.linear_attn.out_proj	2,097,152	264,823,968
model.language_model.layers.0.mlp.down_proj	3,670,016	268,493,984
model.language_model.layers.0.mlp.gate_proj	3,670,016	272,164,000
model.language_model.layers.0.mlp.up_proj	3,670,016	275,834,016
model.language_model.layers.0. post_attention_layernorm	1,024	275,835,040
model.language_model.layers.1.input_layernorm	1,024	275,836,064
model.language_model.layers.1.linear_attn	32	275,836,096
model.language_model.layers.1.linear_attn.conv1d	24,576	275,860,672
model.language_model.layers.1.linear_attn.in_proj_a	16,384	275,877,056
model.language_model.layers.1.linear_attn.in_proj_b	16,384	275,893,440
model.language_model.layers.1.linear_attn. in_proj_qkv	6,291,456	282,184,896
model.language_model.layers.1.linear_attn.in_proj_z	2,097,152	284,282,048
model.language_model.layers.1.linear_attn.norm	128	284,282,176
model.language_model.layers.1.linear_attn.out_proj	2,097,152	286,379,328
model.language_model.layers.1.mlp.down_proj	3,670,016	290,049,344
model.language_model.layers.1.mlp.gate_proj	3,670,016	293,719,360
model.language_model.layers.1.mlp.up_proj	3,670,016	297,389,376
model.language_model.layers.1. post_attention_layernorm	1,024	297,390,400
model.language_model.layers.2.input_layernorm	1,024	297,391,424
model.language_model.layers.2.linear_attn	32	297,391,456
model.language_model.layers.2.linear_attn.conv1d	24,576	297,416,032
model.language_model.layers.2.linear_attn.in_proj_a	16,384	297,432,416
model.language_model.layers.2.linear_attn.in_proj_b	16,384	297,448,800
model.language_model.layers.2.linear_attn. in_proj_qkv	6,291,456	303,740,256
model.language_model.layers.2.linear_attn.in_proj_z	2,097,152	305,837,408
model.language_model.layers.2.linear_attn.norm	128	305,837,536

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Module	Params	Running total
model.language_model.layers.2.linear_attn.out_proj	2,097,152	307,934,688
model.language_model.layers.2.mlp.down_proj	3,670,016	311,604,704
model.language_model.layers.2.mlp.gate_proj	3,670,016	315,274,720
model.language_model.layers.2.mlp.up_proj	3,670,016	318,944,736
model.language_model.layers.2. post_attention_layernorm	1,024	318,945,760
model.language_model.layers.3.input_layernorm	1,024	318,946,784
model.language_model.layers.3.mlp.down_proj	3,670,016	322,616,800
model.language_model.layers.3.mlp.gate_proj	3,670,016	326,286,816
model.language_model.layers.3.mlp.up_proj	3,670,016	329,956,832
model.language_model.layers.3. post_attention_layernorm	1,024	329,957,856
model.language_model.layers.3.self_attn.k_norm	256	329,958,112
model.language_model.layers.3.self_attn.k_proj	524,288	330,482,400
model.language_model.layers.3.self_attn.o_proj	2,097,152	332,579,552
model.language_model.layers.3.self_attn.q_norm	256	332,579,808
model.language_model.layers.3.self_attn.q_proj	4,194,304	336,774,112
model.language_model.layers.3.self_attn.v_proj	524,288	337,298,400
model.language_model.layers.4.input_layernorm	1,024	337,299,424
model.language_model.layers.4.linear_attn	32	337,299,456
model.language_model.layers.4.linear_attn.conv1d	24,576	337,324,032
model.language_model.layers.4.linear_attn.in_proj_a	16,384	337,340,416
model.language_model.layers.4.linear_attn.in_proj_b	16,384	337,356,800
model.language_model.layers.4.linear_attn. in_proj_qkv	6,291,456	343,648,256
model.language_model.layers.4.linear_attn.in_proj_z	2,097,152	345,745,408
model.language_model.layers.4.linear_attn.norm	128	345,745,536
model.language_model.layers.4.linear_attn.out_proj	2,097,152	347,842,688
model.language_model.layers.4.mlp.down_proj	3,670,016	351,512,704
model.language_model.layers.4.mlp.gate_proj	3,670,016	355,182,720
model.language_model.layers.4.mlp.up_proj	3,670,016	358,852,736
model.language_model.layers.4. post_attention_layernorm	1,024	358,853,760
model.language_model.layers.5.input_layernorm	1,024	358,854,784
model.language_model.layers.5.linear_attn	32	358,854,816
model.language_model.layers.5.linear_attn.conv1d	24,576	358,879,392
model.language_model.layers.5.linear_attn.in_proj_a	16,384	358,895,776
model.language_model.layers.5.linear_attn.in_proj_b	16,384	358,912,160
model.language_model.layers.5.linear_attn. in_proj_qkv	6,291,456	365,203,616
model.language_model.layers.5.linear_attn.in_proj_z	2,097,152	367,300,768
model.language_model.layers.5.linear_attn.norm	128	367,300,896
model.language_model.layers.5.linear_attn.out_proj	2,097,152	369,398,048
model.language_model.layers.5.mlp.down_proj	3,670,016	373,068,064
model.language_model.layers.5.mlp.gate_proj	3,670,016	376,738,080
model.language_model.layers.5.mlp.up_proj	3,670,016	380,408,096
model.language_model.layers.5. post_attention_layernorm	1,024	380,409,120
model.language_model.layers.6.input_layernorm	1,024	380,410,144
model.language_model.layers.6.linear_attn	32	380,410,176
model.language_model.layers.6.linear_attn.conv1d	24,576	380,434,752
model.language_model.layers.6.linear_attn.in_proj_a	16,384	380,451,136
model.language_model.layers.6.linear_attn.in_proj_b	16,384	380,467,520
model.language_model.layers.6.linear_attn. in_proj_qkv	6,291,456	386,758,976
model.language_model.layers.6.linear_attn.in_proj_z	2,097,152	388,856,128
model.language_model.layers.6.linear_attn.norm	128	388,856,256
model.language_model.layers.6.linear_attn.out_proj	2,097,152	390,953,408

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Module	Params	Running total
model.language_model.layers.6.mlp.down_proj	3,670,016	394,623,424
model.language_model.layers.6.mlp.gate_proj	3,670,016	398,293,440
model.language_model.layers.6.mlp.up_proj	3,670,016	401,963,456
model.language_model.layers.6. post_attention_layernorm	1,024	401,964,480
model.language_model.layers.7.input_layernorm	1,024	401,965,504
model.language_model.layers.7.mlp.down_proj	3,670,016	405,635,520
model.language_model.layers.7.mlp.gate_proj	3,670,016	409,305,536
model.language_model.layers.7.mlp.up_proj	3,670,016	412,975,552
model.language_model.layers.7. post_attention_layernorm	1,024	412,976,576
model.language_model.layers.7.self_attn.k_norm	256	412,976,832
model.language_model.layers.7.self_attn.k_proj	524,288	413,501,120
model.language_model.layers.7.self_attn.o_proj	2,097,152	415,598,272
model.language_model.layers.7.self_attn.q_norm	256	415,598,528
model.language_model.layers.7.self_attn.q_proj	4,194,304	419,792,832
model.language_model.layers.7.self_attn.v_proj	524,288	420,317,120
model.language_model.layers.8.input_layernorm	1,024	420,318,144
model.language_model.layers.8.linear_attn	32	420,318,176
model.language_model.layers.8.linear_attn.conv1d	24,576	420,342,752
model.language_model.layers.8.linear_attn.in_proj_a	16,384	420,359,136
model.language_model.layers.8.linear_attn.in_proj_b	16,384	420,375,520
model.language_model.layers.8.linear_attn. in_proj_qkv	6,291,456	426,666,976
model.language_model.layers.8.linear_attn.in_proj_z	2,097,152	428,764,128
model.language_model.layers.8.linear_attn.norm	128	428,764,256
model.language_model.layers.8.linear_attn.out_proj	2,097,152	430,861,408
model.language_model.layers.8.mlp.down_proj	3,670,016	434,531,424
model.language_model.layers.8.mlp.gate_proj	3,670,016	438,201,440
model.language_model.layers.8.mlp.up_proj	3,670,016	441,871,456
model.language_model.layers.8. post_attention_layernorm	1,024	441,872,480
model.language_model.layers.9.input_layernorm	1,024	441,873,504
model.language_model.layers.9.linear_attn	32	441,873,536
model.language_model.layers.9.linear_attn.conv1d	24,576	441,898,112
model.language_model.layers.9.linear_attn.in_proj_a	16,384	441,914,496
model.language_model.layers.9.linear_attn.in_proj_b	16,384	441,930,880
model.language_model.layers.9.linear_attn. in_proj_qkv	6,291,456	448,222,336
model.language_model.layers.9.linear_attn.in_proj_z	2,097,152	450,319,488
model.language_model.layers.9.linear_attn.norm	128	450,319,616
model.language_model.layers.9.linear_attn.out_proj	2,097,152	452,416,768
model.language_model.layers.9.mlp.down_proj	3,670,016	456,086,784
model.language_model.layers.9.mlp.gate_proj	3,670,016	459,756,800
model.language_model.layers.9.mlp.up_proj	3,670,016	463,426,816
model.language_model.layers.9. post_attention_layernorm	1,024	463,427,840
model.language_model.layers.10.input_layernorm	1,024	463,428,864
model.language_model.layers.10.linear_attn	32	463,428,896
model.language_model.layers.10.linear_attn.conv1d	24,576	463,453,472
model.language_model.layers.10.linear_attn. in_proj_a	16,384	463,469,856
model.language_model.layers.10.linear_attn. in_proj_b	16,384	463,486,240
model.language_model.layers.10.linear_attn. in_proj_qkv	6,291,456	469,777,696
model.language_model.layers.10.linear_attn. in_proj_z	2,097,152	471,874,848

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Module	Params	Running total
model.language_model.layers.10.linear_attn.norm	128	471,874,976
model.language_model.layers.10.linear_attn.out_proj	2,097,152	473,972,128
model.language_model.layers.10.mlp.down_proj	3,670,016	477,642,144
model.language_model.layers.10.mlp.gate_proj	3,670,016	481,312,160
model.language_model.layers.10.mlp.up_proj	3,670,016	484,982,176
model.language_model.layers.10.post_attention_layernorm	1,024	484,983,200
model.language_model.layers.11.input_layernorm	1,024	484,984,224
model.language_model.layers.11.mlp.down_proj	3,670,016	488,654,240
model.language_model.layers.11.mlp.gate_proj	3,670,016	492,324,256
model.language_model.layers.11.mlp.up_proj	3,670,016	495,994,272
model.language_model.layers.11.post_attention_layernorm	1,024	495,995,296
model.language_model.layers.11.self_attn.k_norm	256	495,995,552
model.language_model.layers.11.self_attn.k_proj	524,288	496,519,840
model.language_model.layers.11.self_attn.o_proj	2,097,152	498,616,992
model.language_model.layers.11.self_attn.q_norm	256	498,617,248
model.language_model.layers.11.self_attn.q_proj	4,194,304	502,811,552
model.language_model.layers.11.self_attn.v_proj	524,288	503,335,840
model.language_model.layers.12.input_layernorm	1,024	503,336,864
model.language_model.layers.12.linear_attn	32	503,336,896
model.language_model.layers.12.linear_attn.conv1d	24,576	503,361,472
model.language_model.layers.12.linear_attn.in_proj_a	16,384	503,377,856
model.language_model.layers.12.linear_attn.in_proj_b	16,384	503,394,240
model.language_model.layers.12.linear_attn.in_proj_qkv	6,291,456	509,685,696
model.language_model.layers.12.linear_attn.in_proj_z	2,097,152	511,782,848
model.language_model.layers.12.linear_attn.norm	128	511,782,976
model.language_model.layers.12.linear_attn.out_proj	2,097,152	513,880,128
model.language_model.layers.12.mlp.down_proj	3,670,016	517,550,144
model.language_model.layers.12.mlp.gate_proj	3,670,016	521,220,160
model.language_model.layers.12.mlp.up_proj	3,670,016	524,890,176
model.language_model.layers.12.post_attention_layernorm	1,024	524,891,200
model.language_model.layers.13.input_layernorm	1,024	524,892,224
model.language_model.layers.13.linear_attn	32	524,892,256
model.language_model.layers.13.linear_attn.conv1d	24,576	524,916,832
model.language_model.layers.13.linear_attn.in_proj_a	16,384	524,933,216
model.language_model.layers.13.linear_attn.in_proj_b	16,384	524,949,600
model.language_model.layers.13.linear_attn.in_proj_qkv	6,291,456	531,241,056
model.language_model.layers.13.linear_attn.in_proj_z	2,097,152	533,338,208
model.language_model.layers.13.linear_attn.norm	128	533,338,336
model.language_model.layers.13.linear_attn.out_proj	2,097,152	535,435,488
model.language_model.layers.13.mlp.down_proj	3,670,016	539,105,504
model.language_model.layers.13.mlp.gate_proj	3,670,016	542,775,520
model.language_model.layers.13.mlp.up_proj	3,670,016	546,445,536
model.language_model.layers.13.post_attention_layernorm	1,024	546,446,560
model.language_model.layers.14.input_layernorm	1,024	546,447,584
model.language_model.layers.14.linear_attn	32	546,447,616
model.language_model.layers.14.linear_attn.conv1d	24,576	546,472,192

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Module	Params	Running total
model.language_model.layers.14.linear_attn.in_proj_a	16,384	546,488,576
model.language_model.layers.14.linear_attn.in_proj_b	16,384	546,504,960
model.language_model.layers.14.linear_attn.in_proj_qkv	6,291,456	552,796,416
model.language_model.layers.14.linear_attn.in_proj_z	2,097,152	554,893,568
model.language_model.layers.14.linear_attn.norm	128	554,893,696
model.language_model.layers.14.linear_attn.out_proj	2,097,152	556,990,848
model.language_model.layers.14.mlp.down_proj	3,670,016	560,660,864
model.language_model.layers.14.mlp.gate_proj	3,670,016	564,330,880
model.language_model.layers.14.mlp.up_proj	3,670,016	568,000,896
model.language_model.layers.14.post_attention_layernorm	1,024	568,001,920
model.language_model.layers.15.input_layernorm	1,024	568,002,944
model.language_model.layers.15.mlp.down_proj	3,670,016	571,672,960
model.language_model.layers.15.mlp.gate_proj	3,670,016	575,342,976
model.language_model.layers.15.mlp.up_proj	3,670,016	579,012,992
model.language_model.layers.15.post_attention_layernorm	1,024	579,014,016
model.language_model.layers.15.self_attn.k_norm	256	579,014,272
model.language_model.layers.15.self_attn.k_proj	524,288	579,538,560
model.language_model.layers.15.self_attn.o_proj	2,097,152	581,635,712
model.language_model.layers.15.self_attn.q_norm	256	581,635,968
model.language_model.layers.15.self_attn.q_proj	4,194,304	585,830,272
model.language_model.layers.15.self_attn.v_proj	524,288	586,354,560
model.language_model.layers.16.input_layernorm	1,024	586,355,584
model.language_model.layers.16.linear_attn	32	586,355,616
model.language_model.layers.16.linear_attn.conv1d	24,576	586,380,192
model.language_model.layers.16.linear_attn.in_proj_a	16,384	586,396,576
model.language_model.layers.16.linear_attn.in_proj_b	16,384	586,412,960
model.language_model.layers.16.linear_attn.in_proj_qkv	6,291,456	592,704,416
model.language_model.layers.16.linear_attn.in_proj_z	2,097,152	594,801,568
model.language_model.layers.16.linear_attn.norm	128	594,801,696
model.language_model.layers.16.linear_attn.out_proj	2,097,152	596,898,848
model.language_model.layers.16.mlp.down_proj	3,670,016	600,568,864
model.language_model.layers.16.mlp.gate_proj	3,670,016	604,238,880
model.language_model.layers.16.mlp.up_proj	3,670,016	607,908,896
model.language_model.layers.16.post_attention_layernorm	1,024	607,909,920
model.language_model.layers.17.input_layernorm	1,024	607,910,944
model.language_model.layers.17.linear_attn	32	607,910,976
model.language_model.layers.17.linear_attn.conv1d	24,576	607,935,552
model.language_model.layers.17.linear_attn.in_proj_a	16,384	607,951,936
model.language_model.layers.17.linear_attn.in_proj_b	16,384	607,968,320
model.language_model.layers.17.linear_attn.in_proj_qkv	6,291,456	614,259,776
model.language_model.layers.17.linear_attn.in_proj_z	2,097,152	616,356,928
model.language_model.layers.17.linear_attn.norm	128	616,357,056
model.language_model.layers.17.linear_attn.out_proj	2,097,152	618,454,208

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Module	Params	Running total
model.language_model.layers.17.mlp.down_proj	3,670,016	622,124,224
model.language_model.layers.17.mlp.gate_proj	3,670,016	625,794,240
model.language_model.layers.17.mlp.up_proj	3,670,016	629,464,256
model.language_model.layers.17. post_attention_layernorm	1,024	629,465,280
model.language_model.layers.18.input_layernorm	1,024	629,466,304
model.language_model.layers.18.linear_attn	32	629,466,336
model.language_model.layers.18.linear_attn.conv1d	24,576	629,490,912
model.language_model.layers.18.linear_attn. in_proj_a	16,384	629,507,296
model.language_model.layers.18.linear_attn. in_proj_b	16,384	629,523,680
model.language_model.layers.18.linear_attn. in_proj_qkv	6,291,456	635,815,136
model.language_model.layers.18.linear_attn. in_proj_z	2,097,152	637,912,288
model.language_model.layers.18.linear_attn.norm	128	637,912,416
model.language_model.layers.18.linear_attn.out_proj	2,097,152	640,009,568
model.language_model.layers.18.mlp.down_proj	3,670,016	643,679,584
model.language_model.layers.18.mlp.gate_proj	3,670,016	647,349,600
model.language_model.layers.18.mlp.up_proj	3,670,016	651,019,616
model.language_model.layers.18. post_attention_layernorm	1,024	651,020,640
model.language_model.layers.19.input_layernorm	1,024	651,021,664
model.language_model.layers.19.mlp.down_proj	3,670,016	654,691,680
model.language_model.layers.19.mlp.gate_proj	3,670,016	658,361,696
model.language_model.layers.19.mlp.up_proj	3,670,016	662,031,712
model.language_model.layers.19. post_attention_layernorm	1,024	662,032,736
model.language_model.layers.19.self_attn.k_norm	256	662,032,992
model.language_model.layers.19.self_attn.k_proj	524,288	662,557,280
model.language_model.layers.19.self_attn.o_proj	2,097,152	664,654,432
model.language_model.layers.19.self_attn.q_norm	256	664,654,688
model.language_model.layers.19.self_attn.q_proj	4,194,304	668,848,992
model.language_model.layers.19.self_attn.v_proj	524,288	669,373,280
model.language_model.layers.20.input_layernorm	1,024	669,374,304
model.language_model.layers.20.linear_attn	32	669,374,336
model.language_model.layers.20.linear_attn.conv1d	24,576	669,398,912
model.language_model.layers.20.linear_attn. in_proj_a	16,384	669,415,296
model.language_model.layers.20.linear_attn. in_proj_b	16,384	669,431,680
model.language_model.layers.20.linear_attn. in_proj_qkv	6,291,456	675,723,136
model.language_model.layers.20.linear_attn. in_proj_z	2,097,152	677,820,288
model.language_model.layers.20.linear_attn.norm	128	677,820,416
model.language_model.layers.20.linear_attn.out_proj	2,097,152	679,917,568
model.language_model.layers.20.mlp.down_proj	3,670,016	683,587,584
model.language_model.layers.20.mlp.gate_proj	3,670,016	687,257,600
model.language_model.layers.20.mlp.up_proj	3,670,016	690,927,616
model.language_model.layers.20. post_attention_layernorm	1,024	690,928,640
model.language_model.layers.21.input_layernorm	1,024	690,929,664
model.language_model.layers.21.linear_attn	32	690,929,696
model.language_model.layers.21.linear_attn.conv1d	24,576	690,954,272
model.language_model.layers.21.linear_attn. in_proj_a	16,384	690,970,656

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Module	Params	Running total
model.language_model.layers.21.linear_attn.in_proj_b	16,384	690,987,040
model.language_model.layers.21.linear_attn.in_proj_qkv	6,291,456	697,278,496
model.language_model.layers.21.linear_attn.in_proj_z	2,097,152	699,375,648
model.language_model.layers.21.linear_attn.norm	128	699,375,776
model.language_model.layers.21.linear_attn.out_proj	2,097,152	701,472,928
model.language_model.layers.21.mlp.down_proj	3,670,016	705,142,944
model.language_model.layers.21.mlp.gate_proj	3,670,016	708,812,960
model.language_model.layers.21.mlp.up_proj	3,670,016	712,482,976
model.language_model.layers.21.post_attention_layernorm	1,024	712,484,000
model.language_model.layers.22.input_layernorm	1,024	712,485,024
model.language_model.layers.22.linear_attn	32	712,485,056
model.language_model.layers.22.linear_attn.conv1d	24,576	712,509,632
model.language_model.layers.22.linear_attn.in_proj_a	16,384	712,526,016
model.language_model.layers.22.linear_attn.in_proj_b	16,384	712,542,400
model.language_model.layers.22.linear_attn.in_proj_qkv	6,291,456	718,833,856
model.language_model.layers.22.linear_attn.in_proj_z	2,097,152	720,931,008
model.language_model.layers.22.linear_attn.norm	128	720,931,136
model.language_model.layers.22.linear_attn.out_proj	2,097,152	723,028,288
model.language_model.layers.22.mlp.down_proj	3,670,016	726,698,304
model.language_model.layers.22.mlp.gate_proj	3,670,016	730,368,320
model.language_model.layers.22.mlp.up_proj	3,670,016	734,038,336
model.language_model.layers.22.post_attention_layernorm	1,024	734,039,360
model.language_model.layers.23.input_layernorm	1,024	734,040,384
model.language_model.layers.23.mlp.down_proj	3,670,016	737,710,400
model.language_model.layers.23.mlp.gate_proj	3,670,016	741,380,416
model.language_model.layers.23.mlp.up_proj	3,670,016	745,050,432
model.language_model.layers.23.post_attention_layernorm	1,024	745,051,456
model.language_model.layers.23.self_attn.k_norm	256	745,051,712
model.language_model.layers.23.self_attn.k_proj	524,288	745,576,000
model.language_model.layers.23.self_attn.o_proj	2,097,152	747,673,152
model.language_model.layers.23.self_attn.q_norm	256	747,673,408
model.language_model.layers.23.self_attn.q_proj	4,194,304	751,867,712
model.language_model.layers.23.self_attn.v_proj	524,288	752,392,000
model.language_model.norm	1,024	752,393,024
model.visual.blocks.0.attn.proj	590,592	752,983,616
model.visual.blocks.0.attn.qkv	1,771,776	754,755,392
model.visual.blocks.0.mlp.linear_fc1	2,362,368	757,117,760
model.visual.blocks.0.mlp.linear_fc2	2,360,064	759,477,824
model.visual.blocks.0.norm1	1,536	759,479,360
model.visual.blocks.0.norm2	1,536	759,480,896
model.visual.blocks.1.attn.proj	590,592	760,071,488
model.visual.blocks.1.attn.qkv	1,771,776	761,843,264
model.visual.blocks.1.mlp.linear_fc1	2,362,368	764,205,632
model.visual.blocks.1.mlp.linear_fc2	2,360,064	766,565,696
model.visual.blocks.1.norm1	1,536	766,567,232
model.visual.blocks.1.norm2	1,536	766,568,768
model.visual.blocks.2.attn.proj	590,592	767,159,360
model.visual.blocks.2.attn.qkv	1,771,776	768,931,136

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Module	Params	Running total
model.visual.blocks.2.mlp.linear_fc1	2,362,368	771,293,504
model.visual.blocks.2.mlp.linear_fc2	2,360,064	773,653,568
model.visual.blocks.2.norm1	1,536	773,655,104
model.visual.blocks.2.norm2	1,536	773,656,640
model.visual.blocks.3.attn.proj	590,592	774,247,232
model.visual.blocks.3.attn.qkv	1,771,776	776,019,008
model.visual.blocks.3.mlp.linear_fc1	2,362,368	778,381,376
model.visual.blocks.3.mlp.linear_fc2	2,360,064	780,741,440
model.visual.blocks.3.norm1	1,536	780,742,976
model.visual.blocks.3.norm2	1,536	780,744,512
model.visual.blocks.4.attn.proj	590,592	781,335,104
model.visual.blocks.4.attn.qkv	1,771,776	783,106,880
model.visual.blocks.4.mlp.linear_fc1	2,362,368	785,469,248
model.visual.blocks.4.mlp.linear_fc2	2,360,064	787,829,312
model.visual.blocks.4.norm1	1,536	787,830,848
model.visual.blocks.4.norm2	1,536	787,832,384
model.visual.blocks.5.attn.proj	590,592	788,422,976
model.visual.blocks.5.attn.qkv	1,771,776	790,194,752
model.visual.blocks.5.mlp.linear_fc1	2,362,368	792,557,120
model.visual.blocks.5.mlp.linear_fc2	2,360,064	794,917,184
model.visual.blocks.5.norm1	1,536	794,918,720
model.visual.blocks.5.norm2	1,536	794,920,256
model.visual.blocks.6.attn.proj	590,592	795,510,848
model.visual.blocks.6.attn.qkv	1,771,776	797,282,624
model.visual.blocks.6.mlp.linear_fc1	2,362,368	799,644,992
model.visual.blocks.6.mlp.linear_fc2	2,360,064	802,005,056
model.visual.blocks.6.norm1	1,536	802,006,592
model.visual.blocks.6.norm2	1,536	802,008,128
model.visual.blocks.7.attn.proj	590,592	802,598,720
model.visual.blocks.7.attn.qkv	1,771,776	804,370,496
model.visual.blocks.7.mlp.linear_fc1	2,362,368	806,732,864
model.visual.blocks.7.mlp.linear_fc2	2,360,064	809,092,928
model.visual.blocks.7.norm1	1,536	809,094,464
model.visual.blocks.7.norm2	1,536	809,096,000
model.visual.blocks.8.attn.proj	590,592	809,686,592
model.visual.blocks.8.attn.qkv	1,771,776	811,458,368
model.visual.blocks.8.mlp.linear_fc1	2,362,368	813,820,736
model.visual.blocks.8.mlp.linear_fc2	2,360,064	816,180,800
model.visual.blocks.8.norm1	1,536	816,182,336
model.visual.blocks.8.norm2	1,536	816,183,872
model.visual.blocks.9.attn.proj	590,592	816,774,464
model.visual.blocks.9.attn.qkv	1,771,776	818,546,240
model.visual.blocks.9.mlp.linear_fc1	2,362,368	820,908,608
model.visual.blocks.9.mlp.linear_fc2	2,360,064	823,268,672
model.visual.blocks.9.norm1	1,536	823,270,208
model.visual.blocks.9.norm2	1,536	823,271,744
model.visual.blocks.10.attn.proj	590,592	823,862,336
model.visual.blocks.10.attn.qkv	1,771,776	825,634,112
model.visual.blocks.10.mlp.linear_fc1	2,362,368	827,996,480
model.visual.blocks.10.mlp.linear_fc2	2,360,064	830,356,544
model.visual.blocks.10.norm1	1,536	830,358,080
model.visual.blocks.10.norm2	1,536	830,359,616
model.visual.blocks.11.attn.proj	590,592	830,950,208
model.visual.blocks.11.attn.qkv	1,771,776	832,721,984
model.visual.blocks.11.mlp.linear_fc1	2,362,368	835,084,352
model.visual.blocks.11.mlp.linear_fc2	2,360,064	837,444,416
model.visual.blocks.11.norm1	1,536	837,445,952
model.visual.blocks.11.norm2	1,536	837,447,488

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Module	Params	Running total
model.visual.merger.linear_fc1	9,440,256	846,887,744
model.visual.merger.linear_fc2	3,146,752	850,034,496
model.visual.merger.norm	1,536	850,036,032
model.visual.patch_embed.proj	1,180,416	851,216,448
model.visual.pos_embed	1,769,472	852,985,920